Workshop #1

Aeroservoelastictiy of Fixed and Rotary Wing Aircraft

Prof. Giuseppe Quaranta, AA 2025/26

1. **Sensitivity to the number of elements.** Consider the Goland Wing model. Verify what is the minimum number of elements in the FEM model that can bring the error in the modal shape approximation, both for bending and torsion, for all the employed modal shapes below 1%.

2. **Convergence of modal frequencies.** Plot the behavior of the error in the modal frequencies of the first 6 proper orthogonal modes with the number of bending N_w and torsional N_θ modal forms used to compute them. Use Richardson expression to estimate the "exact" modal frequencies.

3. **Optimal modal basis.** Estimate what is the minimal set of proper orthogonal modes that should be used to reduce the error in the flutter speed estimate below 1%.

4. **Sensitivity to parameter modifications**. Compute the sensitivity of the flutter speed to:
    a. +/- 2% variation in the structural bending stiffness
    b. +/- 2% variation in the structural torsional stiffness
    c. +/- 2% in the position of the center of gravity along the chord
    d. +/- 2% lift slope coefficient.

5. **Addition of a tip weight.** Add a weight at the wing tip of 80 kg with a pitching moment of inertia about its center of gravity equal to 15 kg m^2. And compute the flutter speed when:
    a. The tip weight CG is coincident with the elastic axis
    b. The tip weight CG is at 5% of the chord
    c. The tip weight CG is at 50% of the chord
    d. Is it possible to move the CG of the tip weight to a position where the flutter speed is incremented by 15% with respect to configuration a)?

6. **Optional task 1: Mach number.** Add the effect of Mach number in your flutter computational model.

7. **Optional task 2: Aileron**. Compute the flutter speed of the Goland wing with a movable surface. Consider the geometry described here and the following properties:
    a. Total mass of the flap m_f = 8.92 kg
    b. Position of the CG of the flap at half of the chord of the flap
    c. Moment of inertia about the CG J_f = 0.09 kg m^2

d. Stiffness of the control surface K_β = 6.48 10^3 N m/rad
e. Compute the control reversal speed